

# Emissions per Capita: Gigatons and Gigapeople

Physics and Mathematics of Sustainable Energy

College of the Atlantic. September 25, 2026

2024 total GHG emissions, territorial, not including LUCF, measured in Gt of CO<sub>2</sub>e.

- World: 54.4
- China: 14.1
- USA: 6.05
- India: 4.26
- Brazil: 2.80
- Mexico: 0.64
- United Kingdom 0.390
- Nigeria: 0.385
- France: 0.338
- Kenya: 0.010
- Costa Rica 0.015

1. What is the per-capita emissions of the US?
2. What is the per-capita emissions of India?
3. Choose at least one other country and calculate its per-capita emissions.

# Carbon Intensities and Capacity Factors

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1. A home in Texas, USA, uses 800 kWh of electricity a month.
  - (a) What are the emissions associated with this electricity?
  - (b) Put this number in perspective.
  - (c) If you have time, compute the emissions for another state or country.
  
2. A 2MW wind turbine has a capacity factor of 0.32.
  - (a) How much energy does this turbine generate in a year?
  - (b) How many Maine homes could this turbine power?
  
3. The Hoover Dam in the Southwestern US has a nameplate capacity of 2080 MW. The average annual energy generated by the power plant from 1947 to 2008 was 4.2 billion kWh. What is the average capacity factor for the Hoover dam for this time period?